Circuit Design and Simulation Lab

PROJECT PROBLEM STATEMENT

Remote Multi-Channel Capacitance Control via Digital Communication and PWM

1. Motivation and Context

Many modern electronic systems require precise, remotely programmable control over a large bank of reactive components — particularly varactor diodes. A varactor diode is a voltage-controlled capacitor: the reverse-bias voltage applied across its junction determines the junction capacitance it presents to the circuit. By individually controlling the bias voltage on each of many varactor diodes, a system can dynamically shape the electrical behavior of an antenna, filter, or phase-shifting network in real time.

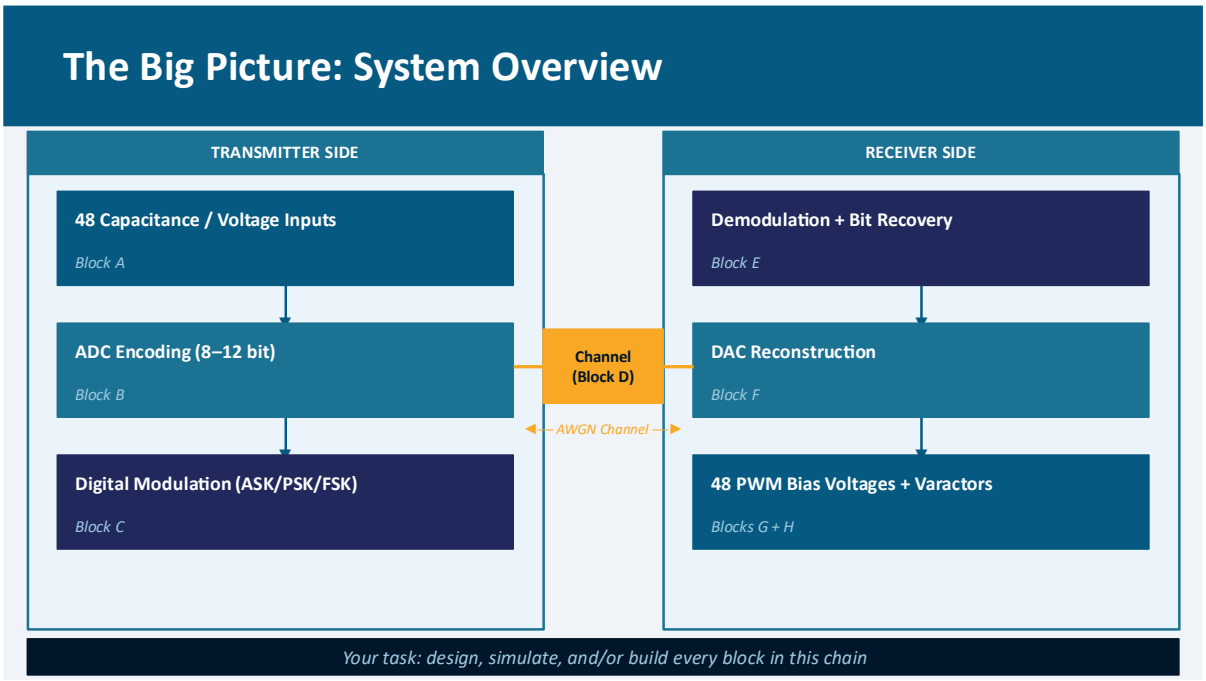
In this project, the transmitter side of a communication link determines what capacitance values are required from an array of 48 varactor diodes at the remote receiver side. This capacitance information must be encoded, transmitted digitally, received, decoded, and finally converted back into the 48 individual DC bias voltages needed to set each varactor diode to its target capacitance.

This end-to-end system integrates four core areas of electronics engineering: analog-to-digital and digital-to-analog conversion, digital modulation and demodulation, microcontroller firmware, and analog circuit design.

2. Problem Statement

Design and implement a complete end-to-end system that:

- **Accepts 48 target capacitance values** (or equivalently, 48 bias voltages) at the transmitter as input.
- **Encodes this information digitally** using an ADC and packages it as a bitstream for transmission.
- **Transmits the bitstream** over a communication channel using a digital modulation scheme of your choice.
- **Receives and demodulates** the signal at the receiver to recover the original bits.
- **Reconstructs the 48 analog bias voltages** using a DAC followed by PWM-based DC generation.
- **Applies these voltages to 48 varactor diodes** and verifies that each diode achieves its target capacitance.



3. System Architecture

Your system must implement the following functional chain. Each block is an independent sub-task that must be designed, justified, and verified.

Block	Block Name	Function	Key Output
A	Transmitter Input	Accept 48 capacitance / voltage values	48 analog voltages (0–V_max)
B	ADC Encoding	Convert analog values to digital bits	Serial bitstream
C	Digital Modulation	Modulate bits onto a carrier signal	Modulated waveform
D	Channel (Simulated)	Represent transmission medium + noise	Received waveform
E	Demodulation	Recover bits from received waveform	Recovered bitstream
F	DAC Reconstruction	Convert bits back to 48 analog voltages	48 analog voltages
G	PWM Voltage Generation	Generate 48 stable DC bias outputs	48 DC bias voltages
H	Varactor Biasing	Apply voltages; measure resulting capacitance	48 varactor C values

4. Block-wise Technical Description

Block A — Transmitter Input Stage

The transmitter holds a lookup table (or receives an external command) specifying the desired capacitance for each of the 48 varactor diodes. Using the varactor's C–V characteristic from its datasheet, the required reverse-bias voltage V_bias for each target capacitance C_target is determined.

- Input: 48 target capacitance values (in pF), e.g., from a predefined codebook table
- Use the varactor datasheet (BB135 or equivalent) to map C_target → V_bias
- These 48 voltages serve as the analog inputs to the ADC stage
- The voltage range for the varactor bias is typically 0–10 V

Block B — ADC Encoding

Each of the 48 bias voltages must be digitized and formatted as a bitstream for transmission. The resolution of digitization determines how accurately the capacitance can be recovered at the receiver.

- Select ADC resolution (minimum 8 bits recommended; justify your choice)
- Calculate the voltage resolution: $\Delta V = V_{ref} / (2^N - 1)$ per channel
- Total bits per update frame = 48 channels × N bits/channel
- Add a frame header (sync word) and CRC checksum for error detection
- Compute the resulting bitstream data rate in kbps

Block C — Digital Modulation

The bitstream must be transmitted using a digital modulation scheme. Students must choose and implement one of the following modulation techniques, and justify the choice based on bandwidth efficiency and complexity.

- Permitted schemes: ASK (OOK), BPSK, BFSK, QPSK, or 8-PSK
- Define the carrier frequency, symbol rate, and spectral bandwidth
- For simulation: generate the modulated waveform in MATLAB/Python or implement on Arduino
- For hardware: use an RF module (e.g., HC-12 FSK module) or SDR-based transmitter
- Derive the theoretical bit error rate (BER) curve for your chosen scheme

Block D — Channel Model

The communication channel introduces noise and possibly multipath. For a simulated system, model the channel as an Additive White Gaussian Noise (AWGN) channel. For hardware implementation, the channel is the physical medium (wired or wireless).

- Simulation: add AWGN noise at specified SNR levels (10 dB, 15 dB, 20 dB)
- Hardware: measure actual received signal power and noise floor
- Compute and plot BER vs. SNR and compare with the theoretical curve

Block E — Demodulation and Bit Recovery

The receiver demodulates the received signal to recover the transmitted bits. This is the inverse of Block C.

- Implement a coherent or non-coherent demodulator matched to your chosen modulation
- Perform frame synchronization using the sync word
- Apply CRC checking to detect bit errors, report frame error rate

- Output: recovered 48 × N bits (the encoded bias voltages)

Block F — DAC Reconstruction

The recovered bits are decoded to reconstruct the 48 analog bias voltages. This is the inverse of Block B.

- Convert each N-bit digital word back to the corresponding analog voltage
- Options: external DAC IC (e.g., MCP4728 quad-DAC, needs 12 ICs for 48 ch), or PWM + RC filter (see Block G)
- Compute quantization error and its effect on the achievable capacitance accuracy
- Measure: DAC linearity, settling time, and output noise

Block G — PWM-Based DC Bias Generation

48 independent DC bias voltages are generated using PWM outputs filtered through RC or RLC low-pass filters. This is the primary voltage generation mechanism for the biasing of the varactor.

- Use a microcontroller (Arduino Mega + PWM driver ICs) to generate 48 independent PWM signals
- Design a low-pass filter (RC or RLC) per channel with cutoff frequency ≤ 10% of PWM frequency
- Target output ripple ≤ 50 mV peak-to-peak on each channel
- If bias range exceeds 5 V (e.g., 0–10 V), include a voltage amplifier stage (gain ≈ 2)
- Verify each channel independently with an oscilloscope or multimeter

Block H — Varactor Characterization and Verification

The final verification step confirms that the bias voltages applied to each varactor diode produce the intended capacitance values.

- Obtain the C–V characteristic of your chosen varactor diode (BB135, SMV1231, or equivalent) from its datasheet
- For hardware: measure capacitance of each varactor using an LCR meter at the applied bias
- For simulation: model the varactor C–V relationship as a table or polynomial and verify in SPICE
- Compute the error between target capacitance and achieved capacitance for all 48 elements
- Tabulate: target C, applied V_bias, measured C, error (pF and %)

6. Deliverables

1. Working Demonstration: Live demo of end-to-end system — transmit a known set of 48 capacitance values and show the resulting bias voltages at the receiver output.
2. Source Code / Simulation Files: Fully commented Arduino firmware, MATLAB/Python scripts, or SPICE/Proteus simulation files (whichever applies).

7. Evaluation Rubric

Assessment Component	Marks	Criteria
ADC/DAC design and analysis (B + F)	5	Resolution justified, quantization error computed
Modulation scheme design (C + E)	5	Correct implementation, BER curve plotted
PWM voltage generation (G)	5	48 channels operational, ripple measured
Varactor characterization (H)	5	C–V curve used; target vs. actual tabulated
System integration and demo (A–H)	10	End-to-end chain functional; error within spec
Report quality	5	Clear, structured, all results included
Viva / Q&A	5	Understanding of each design choice
Total	40	